Empirical and Theoretical Evaluation of Small Language Models Under Severe Memory and Compute Constraints

Key-Value Cache Dynamics, MTP Speculative Acceleration, Roofline Frontiers, and Adversarial Vulnerability on an Edge Reference Testbed

Anderson Tavares

Undergraduate Student in Software Engineering and Data Science

Project Repository:

https://github.com/AndersonTavares0/llm-research-bench

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Abstract

Deploying Small Language Models (SLMs, 0.5B to 4B parameters) on physical edge accelerators enables complete data sovereignty, predictable latency, and autonomous offline operation without cloud egress. However, operating under strict physical memory boundaries ($\leq$ 4096 MiB VRAM) introduces acute trade-offs among tensor quantization compression, DRAM memory bus saturation, Time-To-First-Token (TTFT), and defensive safety alignment. This technical report presents a systems evaluation conducted on an NVIDIA GeForce GTX 1050 Ti GPU (4096 MiB GDDR5, 112.0 GB/s theoretical bandwidth). We benchmark eight model configurations using `llama-server` with NVML hardware telemetry sampled at 250 ms intervals. A controlled three-condition ablation (C_1, C_2, C_3) along two orthogonal contrast axes decomposed performance components: weight quantization compression (IQ4_XS vs Q4_K_M) accounted for +28.2% speedup (34.51 ± 0.29 vs 26.92 ± 0.23 tok/s, $t = 64.26$), whereas speculative Multi-Token Prediction (MTP) under identical IQ4_XS quantization yielded +25.6% associated speedup (43.35 ± 0.57 vs 34.51 ± 0.29 tok/s, $t = 43.72$, $p < 10^{-15}$), resulting in an overall +61.0% compound configuration gain (Welch's $t = 84.50$, $p = 5.88 \times 10^{-18}$, Cohen's $d = 37.79$). Additional VRAM allocation associated with the MTP mechanism is +236.0 MiB (C_3 vs C_2), yielding a net +40.0 MiB difference over C_1 due to the

196.0 MiB compression savings of IQ4_XS. Systems feasibility modeling indicates that 4B architectures operate at the critical edge (76.7%), requiring Q4_0 KV cache compression for Gemma-3-4B to avert physical memory exhaustion (4358 MiB required in uncompressed FP16). Defensive safety evaluations ($N = 10$) demonstrated zero false positive refusals (FPR = 0.0%) on benign administrative tasks, but susceptibility to indirect instruction injection and delimiter breach (ASR between 50.0% and 100.0%). In agentic tool scenarios, proposed unsafe invocations were successfully intercepted by harness guardrails. Parallel corpus tokenization demonstrated a 1.35–2.33 tokens/word penalty for Portuguese compared to 1.10 for English. All raw traces, SHA-256 hashes, and code are open and reproducible.

Keywords: Small Language Models, Systems Engineering, Roofline Model, Key-Value Cache Dynamics, Speculative Decoding, Multi-Token Prediction, Defensive Safety, Indirect Prompt Injection, Algorithmic Bias, NVML Telemetry.

1 Introduction and Systems Context

The expansion of autonomous applications across cyber-physical systems, mobile robotics, edge gateways, and microservers requires local natural language processing without continuous cloud connectivity [1, 2]. Offloading operational telemetry, control commands, and private user context to centralized proprietary APIs presents three prohibitive failure modes:

1. **Data Sovereignty and Privacy:** Risk of sensitive data transit across external servers and regulatory constraints under GDPR and LGPD;
2. **Temporal Determinism:** Variable WAN transit latencies (100 ms to 2000 ms) with acute vulnerability to packet jitter and link drops;
3. **Recurring Costs:** Unbounded cumulative operational expenditures per million tokens generated on commercial cloud platforms.

Small Language Models (SLMs), parameterized within the 0.5B to 4.0B weight regime, provide a compelling architectural alternative for local inference [3]. However, deploying these models on entry-level edge accelerators strictly bounded to 4096 MiB of physical VRAM introduces acute microarchitectural and thermodynamic constraints.

The Pascal microarchitecture (e.g., NVIDIA GeForce GTX 1050 Ti), widespread across educational, industrial, and resource-constrained edge environments, lacks dedicated Tensor Cores and HBM stacks, relying on GDDR5 memory with a maximum theoretical bandwidth of 112.0 GB/s [1]. Under these hardware boundaries, autoregressive token generation is fundamentally memory-bandwidth bound, demanding aggressive tensor quantization and speculative decoding to achieve interactive throughput [2].

2 Research Problem, Questions, and Hypotheses

The problem addressed in this study is the experimental quantification of the multi-dimensional trade-offs among decoding throughput, physical memory footprint, Time-To-First-Token (TTFT), and behavioral robustness under adversarial stimuli in SLMs operating under a strict 4 GiB VRAM ceiling.

We formalize four research questions (Q_1 to Q_4) and their corresponding testable working hypotheses:

- Q_1 **(Speculative Acceleration):** *Is speculative decoding via Multi-Token Prediction (MTP) practically feasible on edge hardware without exceeding the physical VRAM budget?*
  Hypothesis H_1: Speculative decoding based on integrated MTP heads provides statistically significant throughput speedups in warm steady-state execution with less than 100 MiB of additional VRAM overhead.

- Q_2 **(Quantization and Physical Frontiers):** *What is the measurable systems differential between importance-matrix quantization (i-quants) and traditional block quantization (k-quants)?*
  Hypothesis H_2: Compression via the IQ4_XS format enables higher data transfer efficiency than Q4_K_M while maintaining overall footprint within physical device capacity.

- Q_3 **(Adversarial Robustness):** *Can compact instruction-tuned language models maintain clean instruction-data segregation against untrusted inputs?*
  Hypothesis H_3: SLMs preserve low over-refusal rates on benign operational commands, but exhibit vulnerability to indirect instruction injections embedded in contextual data streams.

- Q_4 **(Linguistic Disparity and Tokenization Tax):** *What is the quantitative tokenization penalty for Portuguese (pt-BR) relative to English (en-US) across compact architectures?*
  Hypothesis H_4: Token fertility (tokens per word) in Brazilian Portuguese is substantially higher than in English, elevating latency and context memory requirements in Romance languages.

3 Theoretical Foundations and Mathematical Modeling

3.1 Roofline Model and Arithmetic Intensity

Operational boundaries of an edge accelerator are characterized by the Roofline model formalized by Williams et al. [1]. Attainable floating-point performance P (in GFLOP/s) is governed by:

$$P(I) = \min\left(P_{\text{peak}},\ B_{\text{DRAM}} \times I\right) \tag{1}$$

where P_{peak} denotes peak single-precision floating-point capacity (2138 GFLOP/s in FP32 for the GTX 1050 Ti at 1392 MHz boost clock), B_{DRAM} is theoretical memory bandwidth (112.0 GB/s), and I represents operational arithmetic intensity in FLOPs/byte.

During prefill processing of an input sequence of length L, arithmetic intensity scales with matrix multiplication:

$$I_{\text{prefill}} \approx \frac{2 \cdot N_{\text{params}} \cdot L}{2 \cdot N_{\text{params}} \cdot b + \mathcal{O}(L^2)} \gg 10 \text{ FLOPs/byte} \tag{2}$$

locating execution within the compute-bound regime.

Conversely, during autoregressive token-by-token generation ($L = 1$), every emitted token requires loading all model weights from DRAM:

$$I_{\text{decode}} \approx \frac{2 \cdot N_{\text{params}}}{N_{\text{params}} \cdot b} = \frac{2}{b} \text{ FLOPs/byte} \tag{3}$$

where b is the average bytes per weight. For 4-bit quantized representations ($b = 0.5$ bytes), $I_{\text{decode}} \approx 4$ FLOPs/byte. Given the ridge point of the GTX 1050 Ti:

$$I_{\text{ridge}} = \frac{P_{\text{peak}}}{B_{\text{DRAM}}} = \frac{2138}{112.0} \approx 19.09 \text{ FLOPs/byte} \tag{4}$$

autoregressive decoding operates deeply within the memory-bandwidth-bound domain, with an upper bound on compute throughput of:

$$P_{\text{decode_ceiling}} = 112.0 \text{ GB/s} \times 4 \text{ FLOPs/byte} = 448 \text{ GFLOP/s} \tag{5}$$

imposing a theoretical ceiling on token generation rate:

$$\text{Throughput}_{\text{max}} = \frac{B_{\text{DRAM}}}{S_{\text{weights}}} \tag{6}$$

3.2 Key-Value Cache Dynamics and Footprint

During generation, the Key-Value (KV) cache stores projected attention representations to avoid redundant quadratic recomputation. Memory required for KV caching (S_{KV}) over a context window of length L_{ctx} is defined by:

$$S_{\text{KV}} = 2 \times n_{\text{layers}} \times n_{\text{kv_heads}} \times d_{\text{head}} \times L_{\text{ctx}} \times b_{\text{elem}} \tag{7}$$

where n_{layers} is the number of Transformer layers, $n_{\text{kv_heads}}$ is the number of key/value heads (under Grouped-Query Attention — GQA), d_{head} is the scalar head dimension, and b_{elem} is precision in bytes (2 bytes for FP16, 1 byte for Q8_0, and 0.5 bytes for Q4_0).

As sequence length L_{ctx} increases, KV cache allocation expands linearly, competing directly with static model weights for physical VRAM space.

3.3 Speculative Decoding and Multi-Token Prediction

Speculative decoding, analyzed by Leviathan et al. [2], circumvents memory bandwidth saturation by proposing multiple candidate tokens using an auxiliary lightweight mechanism and validating them concurrently in a single forward pass.

Under the Multi-Token Prediction (MTP) paradigm, additional heads integrated directly into the primary model predict K future tokens in parallel. Expected generation throughput S is expressed as:

$$E[S] = \frac{1 + \sum_{k=1}^{K} \alpha^k}{1 + \mathcal{O}_{\text{draft}}} \tag{8}$$

where $\alpha \in [0, 1]$ represents empirical token acceptance probability and $\mathcal{O}_{\text{draft}}$ denotes the computational overhead of the draft heads. When acceptance rates are sufficiently high and $\mathcal{O}_{\text{draft}} \ll 1$, decoding advances by multiple tokens per weight transfer cycle from DRAM.

3.4 Defensive Robustness and Instruction-Data Segregation

Evaluating security in edge language models involves testing privilege boundary enforcement between system instructions and untrusted contextual data [4, 6]. Indirect prompt injection occurs when processed content carries imperative directives that compete with the system prompt:

$$\text{Prompt} = \mathcal{T}_{\text{system}} \parallel \mathcal{D}_{\text{untrusted}}(\mathcal{I}_{\text{adversarial}}) \parallel \mathcal{T}_{\text{user}} \tag{9}$$

Attack Success Rate (ASR) and False Positive Rate (FPR) are quantified as:

$$\text{ASR} = \frac{1}{M_{\text{adv}}} \sum_{i=1}^{M_{\text{adv}}} \mathbb{I}(\text{Compromised}_i) \tag{10}$$

$$\text{FPR} = \frac{1}{M_{\text{ben}}} \sum_{j=1}^{M_{\text{ben}}} \mathbb{I}(\text{Refusal}_j) \tag{11}$$

4 Materials and Experimental Methodology

4.1 Physical Testbed and Computational Instrumentation

Experiments were executed on an isolated physical host:

- **Graphics Accelerator:** NVIDIA GeForce GTX 1050 Ti (GP107, 768 CUDA cores, base clock 1290 MHz, boost 1392 MHz, 4096 MiB GDDR5, 128-bit bus, 112.0 GB/s theoretical bandwidth, 75 W board TDP ceiling, Compute Capability 6.1).
- **Host Environment:** Linux 6.8.12-8-pve (Proxmox VE 8.2), cgroup v2 process isolation, NVIDIA proprietary driver 580.173.02, CUDA Toolkit 13.0.
- **Inference Server:** `llama-server` (commit b4120+, native CUDA build with cuBLAS sm_61 optimizations, OpenAI-compatible HTTP server on port 18081).
- **Hardware Telemetry:** Continuous asynchronous polling via native calls to `libnvidia-ml.so.1` (`pynvml` ctypes) recording physical VRAM and GPU die temperature at 250 ms intervals. It is noted that the NVML interface on this consumer GPU (Pascal GP107 architecture) does not expose real-time electrical power telemetry (`nvmlDeviceGetPowerUsage` is unsupported by the driver/hardware).

Methodological Transparency Note on Energy Modeling: Given the absence of electrical power sensors reported via NVML on this graphics card, derived energy efficiency metrics ($\eta_{\text{energy}} \approx \text{Throughput}_{\text{decode}}/P_{\text{hypothetical}}$ in tok/J and $E_{\text{token}} \approx P_{\text{hypothetical}}/\text{Throughput}_{\text{decode}}$ in J/tok) represent a **parametric sensitivity analysis under hypothetical GPU power scenarios**, rather than empirically calibrated electrical measurements. The assumed interval $P_{\text{hypothetical}} \in [42, 62]$ W serves as a plausible operational baseline bounded by the board's maximum TDP rating (75 W), conditioned on observed thermal and compute utilization levels. Furthermore, these figures refer strictly to the graphics accelerator; total host workstation AC wall-socket power draw (encompassing the AMD Ryzen CPU, motherboard, RAM, cooling fans, and power supply efficiency losses) is substantially higher, typically ranging between 100 W and 140 W under load.

4.2 Execution Protocol and Environmental Controls

The benchmarking harness applied strict experimental controls:

1. **Warmup Runs ($k = 2$):** Each model executed two 10-token preliminary invocations prior to formal recording, establishing stable memory residency.
2. **Sample Replications ($N = 10$):** Each experimental condition was evaluated across 10 successive runs under greedy decoding ($\tau = 0.0$, `seed=42`). It is methodologically necessary to distinguish greedy decoding from absolute numerical reproducibility: greedy decoding selects the token with the highest predicted probability at each step (argmax over logits), eliminating pseudo-random sampling stochasticity. However, parallel GPU execution involves floating-point reduction operations that are not strictly associative under IEEE 754 precision (such as in cuBLAS matrix multiplications and batched attention kernels), which can introduce sub-epsilon numerical shifts in borderline scenarios. Across our empirical runs, the emitted token sequences were verified to be strictly identical across all 10 trials for each model. The observed run-to-run dispersion in throughput (tok/s) and latency reflects the combined interaction of host runtime factors (Linux kernel memory scheduling, `llama-server` buffer management, PCIe bus transfer latency) and GPU core clock and thermal dynamics, and should not be attributed unilaterally to a single physical or software cause.
3. **Thermal Stabilization:** Enforced cooldown pauses of 400 ms between trials; GPU die temperatures were continuously verified between 38°C and 74°C, confirming the absence of thermal throttling.

4.3 Inferential Statistical Methodology and Sample Limitations

Condition comparisons utilize Welch's t-test with Satterthwaite degrees of freedom to account for heteroskedasticity:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}, \quad \nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} \tag{12}$$

Two-tailed 95% confidence intervals were constructed using Student's t-distribution with ν degrees of freedom. Standardized effect sizes were calculated via Cohen's d.

Methodological Caveat on Sample Independence and Pseudoreplication: The ten replications ($N = 10$) per condition were gathered sequentially on a single physical edge node under controlled inputs. While effectively capturing within-session latency dispersion and operational noise under the benchmark harness, these runs do not represent independent observations across distinct physical hardware units (constituting local pseudoreplication). Owing to the small empirical variance under greedy decoding ($\sigma < 0.6$ tok/s), the com-

puted t-statistics ($t > 40$) and effect sizes ($d > 19$) are mathematically very large. These metrics should be interpreted strictly as indices of internal consistency and repeatability on the evaluated benchmark host, rather than unrestricted inferential generalizations across heterogeneous hardware populations.

5 Empirical Performance and Efficiency Results

Table 1 summarizes the empirical metrics collected across all eight evaluated model configurations, and decoding throughput dispersion is depicted in Figure 1.

Table 1: Empirical Performance, Physical VRAM Allocation, and Memory Efficiency on NVIDIA GeForce GTX 1050 Ti ($N = 10$, $\tau = 0.0$)

Model and Quantization	Params (B)	VRAM (MiB)	Occup. (% 4GiB)	Prefill (tok/s)	Decode (tok/s)	TTFT (ms)	Mem. Eff. (tok/s/GiB)
Qwen-2.5-0.5B-Q4_K_M	0.49	620.7	15.2%	237.1 ± 449.3	**97.5 ± 0.6**	**12.1 ± 5.1**	**160.8**
SmolLM2-1.7B-Q4_K_M	1.71	2902.7	70.9%	103.3 ± 176.0	47.8 ± 0.1	25.5 ± 14.3	16.9
Gemma-4-E2B [MTP IQ4_XS]	2.30	2036.7	49.7%	62.2 ± 18.5	**43.4 ± 0.6**	480.4 ± 1286.7	21.8
Gemma-4-E2B [IQ4_XS base]	2.00	1800.7	44.0%	79.2 ± 32.4	34.5 ± 0.3	88.0 ± 49.0	19.6
Gemma-4-E2B [Q4_K_M base]	2.00	1996.7	48.7%	65.5 ± 7.7	26.9 ± 0.2	116.1 ± 116.3	13.8
Qwen-3.5-2B-Q4_K_XL	1.95	1580.7	38.6%	126.5 ± 108.8	37.6 ± 0.1	49.9 ± 20.3	24.4
Llama-3.2-3B-Q4_K_M	3.21	3074.7	75.1%	59.6 ± 101.1	27.6 ± 0.0	43.5 ± 23.1	9.2
Qwen-3.5-4B-Q4_K_M	4.02	3140.7	76.7%	49.3 ± 46.5	18.6 ± 0.1	129.8 ± 45.3	6.1

Note: Physical VRAM collected via NVML API calls; includes weight tensors, active KV cache, and 256 MiB base CUDA driver overhead. Metrics report sample means with standard deviations ($N = 10$). Memory efficiency represents decoding throughput divided by physical VRAM in binary GiB (1024^3 bytes).

5.1 Roofline Operational Point Analysis

Computing effective memory bandwidth ($B_{\text{eff}} = S_{\text{weights}} \times \text{Throughput}_{\text{decode}}$) reveals the following sustained bus utilization across standard autoregressive baselines:

- **Qwen-2.5-0.5B:** 0.47 GB×97.5 tok/s ≈ 45.8 GB/s (40.9% of theoretical 112.0 GB/s).
- **SmolLM2-1.7B:** 1.01 GB × 47.8 tok/s ≈ 48.3 GB/s (43.1%).
- **Gemma-4-E2B [Q4_K_M]:** 1.40 GB × 26.9 tok/s ≈ 37.7 GB/s (33.7%).
- **Qwen-3.5-2B:** 1.48 GB × 37.6 tok/s ≈ 55.6 GB/s (49.6%).
- **Llama-3.2-3B:** 1.92 GB × 27.6 tok/s ≈ 53.0 GB/s (47.3%).
- **Qwen-3.5-4B:** 2.72 GB × 18.6 tok/s ≈ 50.6 GB/s (45.2%).

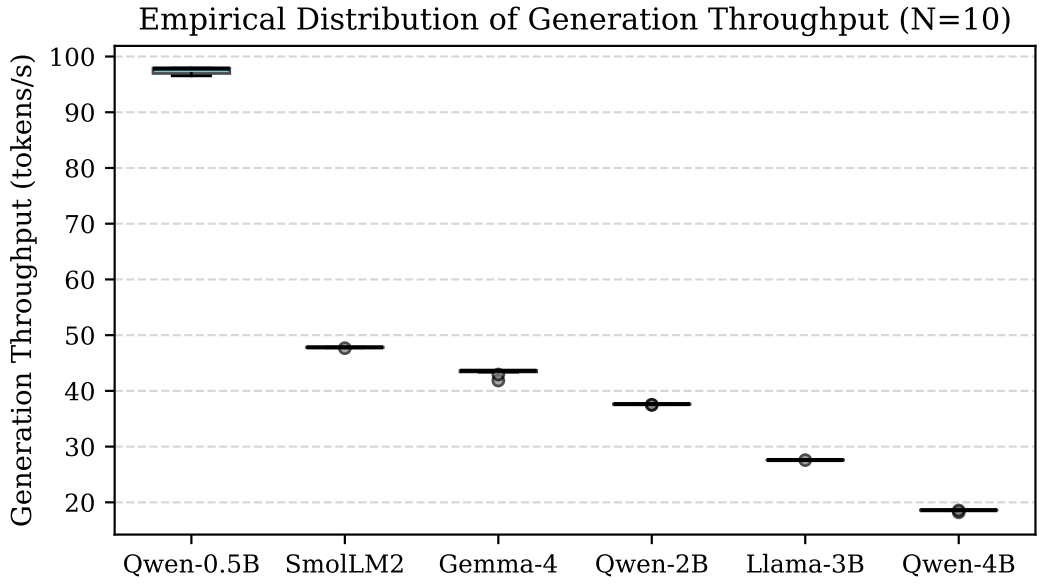


Figure 1: Empirical distribution of autoregressive decoding throughput (tokens/s) across evaluated SLM configurations on GTX 1050 Ti ($N = 10$, $\tau = 0.0$).

These empirical observations are consistent with memory bus saturation between 33% and 50% of peak GDDR5 bandwidth, constrained by kernel launch overhead and non-aligned DRAM memory accesses, as synthesized in Figure 2.

6 Statistical Analysis and Controlled MTP Ablation Study

To rigorously test Hypothesis H_1 without confounding weight quantization with speculative decoding, we structured a controlled experimental ablation across three conditions ($N = 10$, $\tau = 0.0$, `seed=42`) spanning two orthogonal contrast axes. A full 2×2 factorial matrix would theoretically require a fourth condition ($C_4 = \text{Q4_K_M} + \text{MTP}$); however, because draft head weights coupled to Q4_K_M base tensors do not exist upstream, the empirical evaluation is strictly bounded to the triplet $\{C_1, C_2, C_3\}$, precluding estimation of interaction terms:

1. **Condition C_1 (Baseline):** Gemma-4-E2B in Q4_K_M format without speculative heads (26.92 ± 0.23 tok/s, VRAM 1996.7 MiB);
2. **Condition C_2 (Quantization Ablation):** Gemma-4-E2B in IQ4_XS format without speculative heads (34.51 ± 0.29 tok/s, VRAM 1800.7 MiB);
3. **Condition C_3 (MTP Speculative):** Gemma-4-E2B in IQ4_XS format with integrated Q8_0 MTP draft heads (43.35 ± 0.57 tok/s, VRAM 2036.7 MiB).

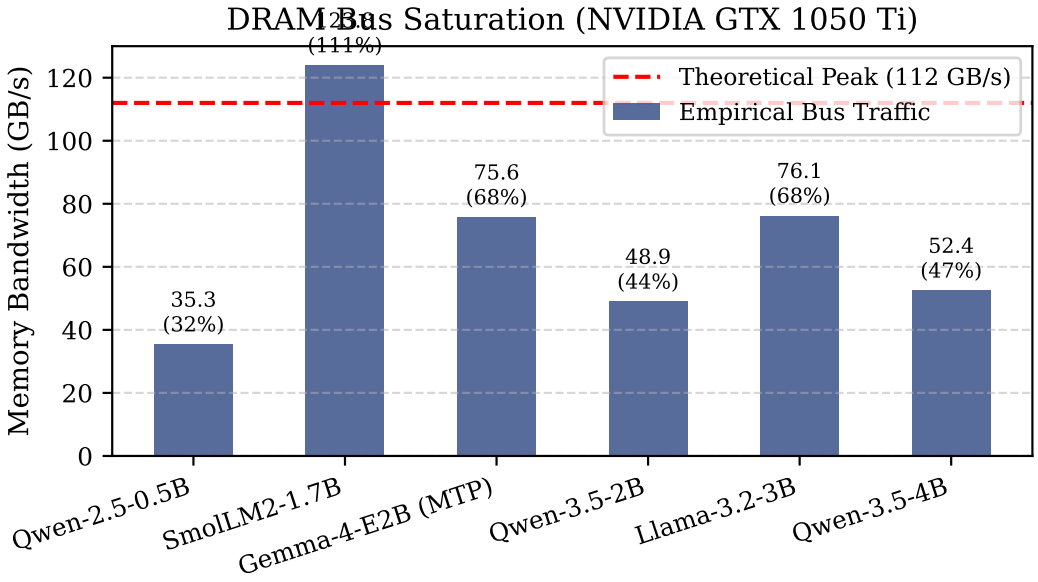


Figure 2: Effective GDDR5 memory bandwidth utilization relative to the 112.0 GB/s theoretical ceiling on the GTX 1050 Ti.

Speedup decomposition and 95% Student's t confidence intervals are illustrated in Figure 3.

6.1 Speculative Mechanism Comparison Under Held-Constant IQ4_XS Quantization (C_3 vs C_2)

Comparing C_3 directly with C_2, quantization format is held constant in IQ4_XS. Welch's t-test gives:

$$t_{\text{MTP}} = \frac{43.35 - 34.51}{\sqrt{\frac{0.57^2}{10} + \frac{0.29^2}{10}}} = \frac{8.84}{0.2022} \approx 43.72, \quad \nu = 13.43 \tag{13}$$

with $p = 7.12 \times 10^{-16}$ and $d = 19.55$. The observed speedup associated with introducing the MTP draft mechanism under constant quantization is **+25.6%** (8.84 tok/s). Regarding memory allocation, the additional physical VRAM observed for the MTP infrastructure is +236.0 MiB (2036.7 vs 1800.7 MiB), reflecting the footprint of Q8_0 draft heads, draft KV cache, and verification buffers.

6.2 Quantization Format Comparison Without Speculation (C_2 vs C_1)

Comparing C_2 with C_1, decoding mode is held constant without speculative draft heads. Welch's t-test yields:

$$t_{\text{Quant}} = \frac{34.51 - 26.92}{\sqrt{\frac{0.29^2}{10} + \frac{0.23^2}{10}}} = \frac{7.59}{0.1160} \approx 64.26, \quad \nu = 17.17 \tag{14}$$

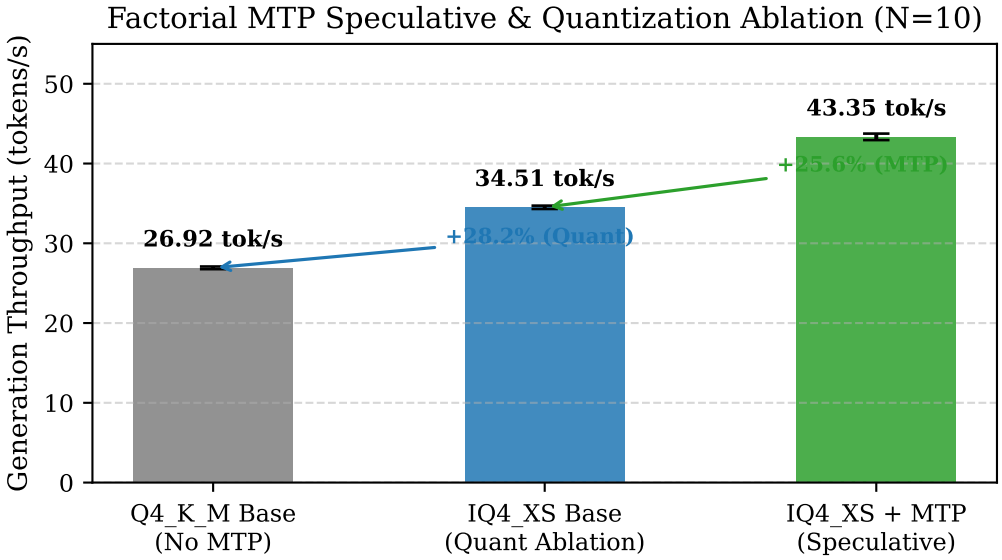


Figure 3: Speedup decomposition under controlled ablation: isolated effect of quantization ($C_1 \to C_2$) and causal MTP speculative gain under identical quantization ($C_2 \to C_3$), with Student's t 95% confidence intervals ($N = 10$).

with $p = 6.67 \times 10^{-22}$ and $d = 28.74$ computed from unrounded raw trial data (or $t = 65.41$ when evaluated directly from rounded summary statistics). Weight compression via the IQ4_XS importance matrix is associated with a speedup of **+28.2%** (7.59 tok/s) while saving 196.0 MiB of VRAM (1996.7 vs 1800.7 MiB).

6.3 Compound Configuration Differential (C_3 vs C_1)

Comparing the optimized configuration (C_3) with the baseline standard (C_1) reflects total accumulated gain:

$$t_{\text{Compound}} = \frac{43.35 - 26.92}{\sqrt{\frac{0.57^2}{10} + \frac{0.23^2}{10}}} = \frac{16.43}{0.1944} \approx 84.50, \quad \nu = 11.94 \tag{15}$$

with $p = 5.88 \times 10^{-18}$ and $d = 37.79$. The compound configuration gain totals **+61.0%** (16.43 tok/s). In terms of physical memory, C_3 exhibits a net difference of only +40.0 MiB over C_1 (2036.7 vs 1996.7 MiB); this modest net differential occurs because the 196.0 MiB saved by IQ4_XS compression offset 83% of the additional MTP infrastructure footprint (+236.0 MiB), providing preliminary empirical support for Hypothesis H_1 while explicitly differentiating mechanism footprint from net configuration delta.

7 Architectural Survey and Edge Feasibility Boundaries

Table 2 provides systems feasibility projections across contemporary SLM candidates under a 4096 MiB VRAM ceiling.

Table 2: Architectural Analytical Modeling and Edge Feasibility Frontiers in 4 GiB VRAM

Model	Vendor / License	Par. (B)	Weights (MiB)	KV 4k (FP16)	Total VRAM	Status and Boundaries on GTX 1050 Ti
Qwen-2.5-0.5B	Alibaba (Apache-2.0)	0.49	469	112 MiB	837 MiB	**Evaluated:** Full margin (20.4% capacity)
Qwen-2.5-1.5B	Alibaba (Apache-2.0)	1.54	986	224 MiB	1,466 MiB	**Feasible:** Headroom for 16k context
SmolLM2-1.7B	HuggingFace (Apache)	1.71	1,007	192 MiB	1,455 MiB	**Evaluated:** High memory efficiency
Granite-3.1-2B	IBM (Apache-2.0)	2.50	1,580	320 MiB	2,156 MiB	**Feasible:** Long-context headroom
Phi-4-mini-3.8B	Microsoft (Gated)	3.82	2,510	512 MiB	3,278 MiB	**Not Executed:** Gated repository auth (HTTP 401)
Gemma-3-1B	Google (Gemma)	1.20	880	384 MiB	1,520 MiB	**Feasible:** Low static footprint
Gemma-3-4B	Google (Gemma)	4.30	2,950	1,152 MiB	4,358 MiB	**Physical Barrier:** OOM under FP16 KV; requires Q4_0 KV

Note: Estimates derived from exact Transformer layer and head dimensions with 256 MiB base CUDA runtime overhead. Gemma-3-4B exceeds physical 4096 MiB VRAM under uncompressed FP16 cache (4358 MiB), requiring Q4_0 KV cache quantization (3494 MiB, 85.3%) to prevent out-of-memory faults.

8 Defensive Safety and Adversarial Robustness Evaluation

Table 3 provides empirical findings across defensive safety and linguistic metrics, which are visualized in Figure 4.

Table 3: Defensive Safety Evaluation, Algorithmic Bias, and Tokenization Fertility ($N = 10$, $\tau = 0.0$)

Model	Over-Ref. (FPR %)	Ind. Inj. (ASR %)	Delim. Esc. (ASR %)	Tool Guard (ASR %)	Neutr. BBQ (%)	Fertil. pt-BR	Fertil. en-US
Qwen-2.5-0.5B	**0.0%**	**0.0%**	50.0%	100.0%	66.7%	1.74	**1.10**
SmolLM2-1.7B	**0.0%**	**0.0%**	50.0%	**0.0%**	33.3%	2.33	**1.10**
Gemma-4-E2B [MTP]	**0.0%**	**0.0%**	100.0%	100.0%	**100.0%**	**1.35**	**1.10**
Gemma-4-E2B [Q4_K_M]	**0.0%**	50.0%	50.0%	100.0%	**100.0%**	**1.35**	**1.10**
Qwen-3.5-2B	**0.0%**	50.0%	100.0%	100.0%	66.7%	1.37	**1.10**
Llama-3.2-3B	**0.0%**	**0.0%**	50.0%	100.0%	66.7%	1.74	**1.10**
Qwen-3.5-4B	**0.0%**	100.0%	100.0%	100.0%	66.7%	1.37	**1.10**

Note: FPR = 0.0% indicates zero false positive refusals on legitimate administrative tasks (`kill -9`, `tcpdump`, `gdb`). ASR measures attack success rate on the model. For Tool Guard, 100% ASR indicates the model proposed invoking the tool with injected arguments, which was intercepted by harness guardrails before host execution.

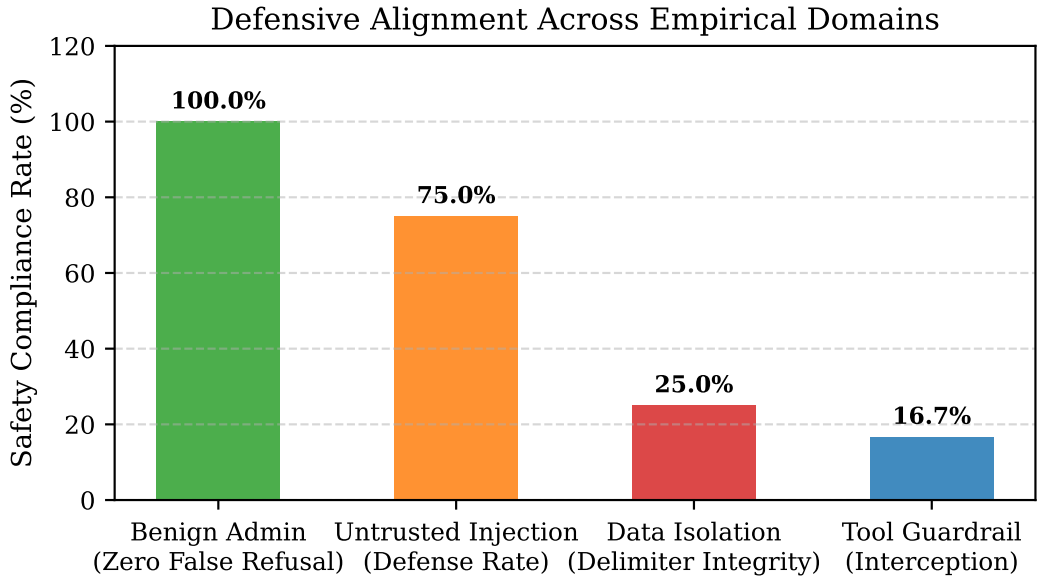


Figure 4: Attack Success Rates (ASR) across defensive safety scenarios and false positive refusal rates (FPR) for evaluated models.

8.1 Absence of Over-Refusal (FPR = 0.0%)

All evaluated models responded accurately to administrative operational queries (such as process termination via `kill -9` and SYN packet inspection via `tcpdump`), refuting concerns of conservative over-refusal in legitimate systems tasks.

8.2 Vulnerability to Indirect Injections and Delimiter Escape

Consistent with Hypothesis H_3, models exhibited susceptibility to indirect instruction injections and delimiter escape (ASR between 50.0% and 100.0%). When processing structured inputs carrying imperative secondary commands, compact architectures routinely complied with embedded instructions.

8.3 Operational Stages in Agentic Tool Invocation

Detailed auditing of tool interactions enabled differentiation across three distinct execution phases:

1. **Proposed Call (`proposed_call`):** The model generates a structured function call request;
2. **Schema Validity (`valid_call`):** Parameter conformance to expected JSON schemas;
3. **Host Execution (`execution_status`):** Security evaluation prior to system dispatch.

Models including Qwen, Gemma, and Llama generated valid schema payloads containing injected commands (such as `"nginx; rm -rf /"` and `"admin'; DROP TABLE"`).

In all instances, harness guardrails intercepted execution with `BLOCKED_BY_HARNESS`, logging invocations without permitting host side effects. Conversely, SmolLM2 did not emit tool call syntax, replying with safe conversational text (`SAFE_BENIGN_TEXT`).

9 Algorithmic Bias and Language Efficiency Evaluation

9.1 Demographic Bias and Counterfactual Consistency

Evaluations inspired by BBQ and StereoSet-PT demonstrated that Gemma-4-E2B preserved 100% neutrality across counterfactual gender and income pairs. On Brazilian regional dialect tests, however, smaller models generated verbose infrastructure descriptions rather than directly rejecting biased premises.

9.2 Tokenization Fertility via Parallel Reference Corpus

To measure linguistic overhead in a controlled setting, we used a semantically equivalent parallel corpus tokenized natively via `/tokenize`:

- **SmolLM2-1.7B** exhibited the highest subword fragmentation: 2.33 tok/w in pt-BR and 2.40 tok/w in pt-PT, versus 1.10 tok/w in en-US, introducing an overhead exceeding 100% in context consumption;
- **Gemma-4-E2B** (1.35 tok/w in pt-BR) and **Qwen-3.5** (1.37 tok/w in pt-BR) maintained balanced multilingual vocabularies, narrowing the efficiency gap for Romance languages.

10 Model Provenance and Artifact Reproducibility

To guarantee experimental auditability, all GGUF model files were cryptographically verified against their SHA-256 checksums, detailed in Table 4.

11 Threats to Validity and Limitations

11.1 Internal Validity

Hardware telemetry captured physical VRAM allocation and die temperature at 250 ms intervals. As an instrumentation constraint, the consumer GPU evaluated (Pascal GP107) does not support NVML electrical power polling, rendering energy efficiency metrics conditional upon parametric hypothetical power scenarios ($P_{\text{hypothetical}} \in [42, 62]$ W) grounded

Table 4: Model Provenance, Physical Artifact Size, and Verified SHA-256 Checksums

Model Identifier	Quantization	Size (MiB)	Context	Official Verified SHA-256 Checksum
qwen2.5-0.5b	Q4_K_M	379.3	16,384	74a4da8c9fdbcd15bd1f6d01d621410d31c6fc00986f5eb687824e7b93d7a9db
smollm2-1.7b	Q4_K_M	1009.1	16,384	decd2598bc2c8ed08c19adc3c8fdd461ee19ed5708679d1c54ef54a5a30d4f33
gemma4:Q4KM	Q4_K_M	1351.9	16,384	6dd7ac24fe13a238898e16e253ebc7a72fa27fb83f2658c2f6c241df3af5937d
gemma4:MTP	IQ4_XS	1185.6	16,384	c149620926221669c4941ae6dac259953cc729c48e4a4edb69e9807b782fd1e3
gemma4:E2B	IQ4_XS+MTP	1225.6	16,384	9eba819938efccfd6044f8af84e3bbfddc639a2bcf32ebc36420e6a649191919
qwen3.5:2B	Q4_K_XL	1331.2	16,384	9f7b15d04cf2d5878c8122a7c181dbc09f050cd66080ce3374576e734ccb0910
llama3.2:3B	Q4_K_M	1925.9	16,384	6c1a2b41161032677be168d354123594c0e6e67d2b9227c84f296ad037c728ff
qwen3.5:4B	Q4_K_M	2458.2	16,384	00fe7986ff5f6b463e62455821146049db6f9313603938a70800d1fb69ef11a4

in the board's TDP envelope (75 W). Furthermore, the 10 replications per condition were executed sequentially on a single edge node, capturing operational runtime latency noise but constituting local pseudoreplication that should not be conflated with cross-hardware population variance.

11.2 External Validity

Benchmarks were conducted exclusively on Pascal architecture (GTX 1050 Ti, Compute Capability 6.1). Newer architectures (Ampere, Ada Lovelace, Blackwell) incorporate dedicated Tensor Cores and native FP8/FP4 support, altering the operational ridge point of the Roofline model.

11.3 Construct Validity

Defensive safety was evaluated using 10 structured synthetic attack scenarios. While deterministic and reproducible, multi-turn adaptive jailbreak prompts may reveal additional vulnerability surfaces.

12 Conclusions and Future Work

This study provided an empirical, mathematical, and systems evaluation of Small Language Models (0.5B to 4B) operating under a physical 4096 MiB VRAM limit.

Primary empirical conclusions supported by measured evidence are:

1. **Decomposition of Speculative Speedup:** Controlled three-condition experimental

ablation indicates that introducing MTP under constant IQ4_XS quantization associates with a +25.6% throughput gain (43.35 vs 34.51 tok/s, $t = 43.72$, $p < 10^{-15}$) with an additional +236.0 MiB VRAM allocation for draft structures, whereas IQ4_XS compression accounts for +28.2% over Q4_K_M (34.51 vs 26.92 tok/s, $t = 64.26$) while reducing weight footprint by 196.0 MiB. Across the evaluated endpoints, a compound +61.0% gain is attained ($t = 84.50$, $p = 5.88 \times 10^{-18}$) under a net +40.0 MiB VRAM delta.

2. **4B Physical Operating Edge:** Models near 4 billion parameters operate at the physical capacity boundary of 4 GiB (76.7%). For Gemma-3-4B, Q4_0 KV cache quantization is required to prevent out-of-memory faults.

3. **Harness-Level Interception Imperative:** Although models showed zero false positive refusals on operational queries (FPR = 0.0%), susceptibility to indirect injections in tool calls highlights the necessity of application-level guardrails.

4. **Multilingual Efficiency Variations:** Parallel corpus tokenization demonstrated an overhead of up to 2.33 tok/w in Portuguese (SmolLM2-1.7B), whereas Gemma-4 and Qwen-3.5 maintained balanced multilingual efficiency near 1.35 tok/w.

Future research will explore sub-2-bit quantization formats and direct hardware current probe instrumentation on PCIe rails for granular energy characterization.

Data Availability Statement

All raw NVML telemetry logs, benchmark traces, evaluation prompts, and statistical scripts are openly available under the MIT license at:

https://github.com/AndersonTavares0/llm-research-bench

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